

MTH253 – Calculus III Cheat Sheet

Chapters 1 & 2: Multivariable Differentiation & Multiple Integration — VNU-HCM University of Science

1. Vectors & 3D Coordinate Geometry

Coordinate Systems & Distance

- Distance in $\mathbb{R}^3$:** The distance between $P_1(x_1, y_1, z_1)$ and $P_2(x_2, y_2, z_2)$ is:

$$d = \sqrt{(x_2 - x_1)^2 + (y_2 - y_1)^2 + (z_2 - z_1)^2}$$

- Equation of a Sphere:** With center $C(h, k, l)$ and radius R :

$$(x - h)^2 + (y - k)^2 + (z - l)^2 = R^2$$

Vectors, Magnitude, and Unit Vectors

- Vector from A to B :** For $A(a_1, b_1, c_1)$ and $B(a_2, b_2, c_2)$:

$$\vec{AB} = \langle a_2 - a_1, b_2 - b_1, c_2 - c_1 \rangle$$

- Magnitude:** For vector $\vec{u} = \langle a, b, c \rangle$:

$$\|\vec{u}\| = \sqrt{a^2 + b^2 + c^2}$$

- Unit Vector:** Vector of length 1 in the direction of $\vec{v} \neq \vec{0}$:

$$\vec{u} = \frac{\vec{v}}{\|\vec{v}\|}$$

- Basis Vectors:** $\vec{i} = \langle 1, 0, 0 \rangle$, $\vec{j} = \langle 0, 1, 0 \rangle$, $\vec{k} = \langle 0, 0, 1 \rangle$.

Dot Product (Scalar Product)

- Definition:** $\vec{u} \cdot \vec{v} = u_1v_1 + u_2v_2 + u_3v_3 = \|\vec{u}\|\|\vec{v}\|\cos\theta$.
- Orthogonality:** $\vec{u} \perp \vec{v} \iff \vec{u} \cdot \vec{v} = 0$.
- Properties:** $\vec{u} \cdot \vec{u} = \|\vec{u}\|^2$; $\vec{u} \cdot \vec{v} = \vec{v} \cdot \vec{u}$.

Cross Product (Vector Product)

- Definition** (Defined only in $\mathbb{R}^3$):

$$\vec{u} \times \vec{v} = \begin{vmatrix} \vec{i} & \vec{j} & \vec{k} \\ u_1 & u_2 & u_3 \\ v_1 & v_2 & v_3 \end{vmatrix} = \langle u_2v_3 - u_3v_2, u_3v_1 - u_1v_3, u_1v_2 - u_2v_1 \rangle$$

- Geometry:** $\vec{u} \times \vec{v}$ is perpendicular to both $\vec{u}$ and $\vec{v}$.
- Magnitude:** $\|\vec{u} \times \vec{v}\| = \|\vec{u}\|\|\vec{v}\|\sin\theta$ (Area of the parallelogram formed by $\vec{u}$ and $\vec{v}$).
- Parallelism:** $\vec{u} \parallel \vec{v} \iff \vec{u} \times \vec{v} = \vec{0}$.

Lines and Planes in Space

- Line L** through $P_0(x_0, y_0, z_0)$ parallel to direction vector $\vec{v} = \langle a, b, c \rangle$:
 - Vector:** $\vec{r}(t) = \vec{r}_0 + t\vec{v}$ or $\langle x, y, z \rangle = \langle x_0, y_0, z_0 \rangle + t\langle a, b, c \rangle$
 - Parametric:** $x = x_0 + at$, $y = y_0 + bt$, $z = z_0 + ct$
 - Symmetric:** $\frac{x-x_0}{a} = \frac{y-y_0}{b} = \frac{z-z_0}{c}$ (for $a, b, c \neq 0$)
- Plane M** through $P_0(x_0, y_0, z_0)$ with normal vector $\vec{n} = \langle a, b, c \rangle$:
 - Vector:** $\vec{n} \cdot (\vec{r} - \vec{r}_0) = 0$
 - Scalar:** $a(x - x_0) + b(y - y_0) + c(z - z_0) = 0$
 - Linear:** $ax + by + cz + d = 0$

2. Limits and Continuity

Functions of Several Variables

- Level Curves:** The set of points where $f(x, y) = k$.
- Level Surfaces:** The set of points where $f(x, y, z) = k$.

Limits

- $\lim_{(x,y) \rightarrow (a,b)} f(x, y) = L \iff \forall \varepsilon > 0, \exists \delta > 0$ such that:

$$0 < \sqrt{(x - a)^2 + (y - b)^2} < \delta \implies |f(x, y) - L| < \varepsilon$$

- Non-existence Test:** If $f(x, y) \rightarrow L_1$ along path C_1 and $f(x, y) \rightarrow L_2$ along path C_2 as $(x, y) \rightarrow (a, b)$, and $L_1 \neq L_2$, then the limit **does not exist**.
- Standard paths to test as $(x, y) \rightarrow (0, 0)$:
 - Axes: $x = 0$ or $y = 0$.
 - Lines: $y = mx$ (shows m -dependence).
 - Parabolas: $y = kx^2$ or $x = ky^2$ (for terms like $x^4 + y^2$).

Squeeze Theorem

If $g(x, y) \leq f(x, y) \leq h(x, y)$ near (a, b) (except possibly at (a, b)) and $\lim_{(x,y) \rightarrow (a,b)} g(x, y) = \lim_{(x,y) \rightarrow (a,b)} h(x, y) = L$, then:

$$\lim_{(x,y) \rightarrow (a,b)} f(x, y) = L$$

Continuity

- f is continuous at (a, b) if $\lim_{(x,y) \rightarrow (a,b)} f(x, y) = f(a, b)$.
- Polynomials, rational, exponential, logarithmic, trigonometric, and root functions are continuous on their domains.

3. Partial Derivatives

First-Order Partial Derivatives

- Respect to x :** Regard y as a constant:

$$f_x(a, b) = \lim_{h \rightarrow 0} \frac{f(a + h, b) - f(a, b)}{h} = \frac{\partial f}{\partial x}$$

- Respect to y :** Regard x as a constant:

$$f_y(a, b) = \lim_{h \rightarrow 0} \frac{f(a, b + h) - f(a, b)}{h} = \frac{\partial f}{\partial y}$$

Higher-Order Derivatives

- Second partials: $f_{xx} = \frac{\partial^2 f}{\partial x^2}$, $f_{yy} = \frac{\partial^2 f}{\partial y^2}$.
- Mixed partials: $f_{xy} = (f_x)_y = \frac{\partial^2 f}{\partial y \partial x}$, $f_{yx} = (f_y)_x = \frac{\partial^2 f}{\partial x \partial y}$.

Clairaut's Theorem

If f is defined on a disk D containing (a, b) , and mixed partials f_{xy} and f_{yx} are both continuous on D , then:

$$f_{xy}(a, b) = f_{yx}(a, b)$$

Tangent Planes & Linearization

- Tangent Plane:** To surface $z = f(x, y)$ at $P(x_0, y_0, z_0)$:

$$z - z_0 = f_x(x_0, y_0)(x - x_0) + f_y(x_0, y_0)(y - y_0)$$

- Linearization:** $L(x, y)$ of $f(x, y)$ at (a, b) :

$$L(x, y) = f(a, b) + f_x(a, b)(x - a) + f_y(a, b)(y - b)$$

- Linear Approximation:** $f(x, y) \approx L(x, y)$ for $(x, y) \approx (a, b)$.

Differentiability & Differentials

- Sufficient Condition:** If f_x and f_y exist near (a, b) and are continuous at (a, b) , then f is **differentiable** at (a, b) .
- Total Differential:** For $z = f(x, y)$:

$$dz = f_x(x, y)dx + f_y(x, y)dy = \frac{\partial z}{\partial x}dx + \frac{\partial z}{\partial y}dy$$

We approximate $\Delta z \approx dz$ for small $\Delta x = dx$ and $\Delta y = dy$.

4. Chain Rule & Implicit Differentiation

The Chain Rule

- Case 1:** $z = f(x, y)$ differentiable, $x = g(t)$, $y = h(t)$ differentiable:

$$\frac{dz}{dt} = \frac{\partial z}{\partial x} \frac{dx}{dt} + \frac{\partial z}{\partial y} \frac{dy}{dt}$$

- Case 2:** $z = f(x, y)$, $x = g(s, t)$, $y = h(s, t)$:

$$\frac{\partial z}{\partial s} = \frac{\partial z}{\partial x} \frac{\partial x}{\partial s} + \frac{\partial z}{\partial y} \frac{\partial y}{\partial s}, \quad \frac{\partial z}{\partial t} = \frac{\partial z}{\partial x} \frac{\partial x}{\partial t} + \frac{\partial z}{\partial y} \frac{\partial y}{\partial t}$$

- General Rule:** For $u = f(x_1, \dots, x_n)$ and $x_j = g_j(t_1, \dots, t_m)$:

$$\frac{\partial u}{\partial t_i} = \sum_{j=1}^n \frac{\partial u}{\partial x_j} \frac{\partial x_j}{\partial t_i}$$

Implicit Differentiation

- Single Variable:** If $F(x, y) = 0$ defines $y = f(x)$ implicitly:

$$\frac{dy}{dx} = -\frac{F_x}{F_y}$$

- Multivariable:** If $F(x, y, z) = 0$ defines $z = f(x, y)$ implicitly:

$$\frac{\partial z}{\partial x} = -\frac{F_x}{F_z}, \quad \frac{\partial z}{\partial y} = -\frac{F_y}{F_z}$$

5. Gradients & Directional Derivatives

The Gradient Vector

- For $f(x, y)$, gradient $\nabla f(x, y)$ is:

$$\nabla f(x, y) = \langle f_x(x, y), f_y(x, y) \rangle = f_x \vec{i} + f_y \vec{j}$$

- For $f(x, y, z)$, gradient $\nabla f(x, y, z)$ is:

$$\nabla f(x, y, z) = \langle f_x, f_y, f_z \rangle = f_x \vec{i} + f_y \vec{j} + f_z \vec{k}$$

Directional Derivative

- Rate of change of f in the direction of a **unit vector** $\vec{u} = \langle a, b \rangle$:

$$D_{\vec{u}}f(x, y) = f_x(x, y)a + f_y(x, y)b = \nabla f(x, y) \cdot \vec{u}$$

- If angle θ defines $\vec{u} = \langle \cos\theta, \sin\theta \rangle$:

$$D_{\vec{u}}f(x, y) = f_x(x, y)\cos\theta + f_y(x, y)\sin\theta$$

- Important:** If given direction is $\vec{v}$, you **must** normalize: $\vec{u} = \frac{\vec{v}}{\|\vec{v}\|}$.

Properties of the Gradient

- **Maximum Rate of Change:** Max value of $D_{\vec{u}}f(x, y)$ is $\|\nabla f(x, y)\|$. Occurs when $\vec{u}$ has the same direction as ∇f (i.e. $\vec{u} = \frac{\nabla f}{\|\nabla f\|}$).
- **Minimum Rate of Change:** Min value is $-\|\nabla f(x, y)\|$ in the direction of $-\nabla f$.
- **Orthogonality:** $\nabla f(x_0, y_0)$ is orthogonal to the level curve $f(x, y) = k$ at (x_0, y_0) .

6. Extrema of Multivariable Functions

Local Extrema & Critical Points

- **Critical Point:** (a, b) is a critical point of f if $f_x(a, b) = 0$ and $f_y(a, b) = 0$, or if either partial derivative does not exist.
- If f has a local extremum at (a, b) , then (a, b) is a critical point.

Second Derivatives Test

Let (a, b) be a critical point of f where second partials are continuous. Let $D = D(a, b)$ be:

$$D = f_{xx}(a, b)f_{yy}(a, b) - [f_{xy}(a, b)]^2$$

a) If $D > 0$ and $f_{xx}(a, b) > 0 \implies f(a, b)$ is a **local minimum**.
b) If $D > 0$ and $f_{xx}(a, b) < 0 \implies f(a, b)$ is a **local maximum**.
c) If $D < 0 \implies (a, b)$ is a **saddle point**.
d) If $D = 0 \implies$ **Inconclusive** (need other methods).

Absolute Extrema on Closed Bounded Sets

- **Extreme Value Theorem:** Continuous f on closed, bounded $D \subset \mathbb{R}^2$ attains an absolute max and min.
- **Closed Interval Method:**
 1. Find values of f at critical points of f in D .
 2. Find extreme values of f on the boundary of D .
 3. Compare: largest is absolute max, smallest is absolute min.

Lagrange Multipliers

- To find extrema of $f(x, y, z)$ subject to constraint $g(x, y, z) = k$ (assuming $\nabla g \neq \vec{0}$):
- **Method:** Solve the system:

$$\begin{cases} \nabla f(x, y, z) = \lambda \nabla g(x, y, z) \\ g(x, y, z) = k \end{cases} \iff \begin{cases} f_x = \lambda g_x \\ f_y = \lambda g_y \\ f_z = \lambda g_z \\ g(x, y, z) = k \end{cases}$$

λ is the Lagrange multiplier. Evaluate f at all solutions.

7. Double Integrals

Iterated Integrals (Fubini's Theorem)

- If f is continuous on rectangle $R = [a, b] \times [c, d]$:

$$\iint_R f(x, y) \, dA = \int_a^b \int_c^d f(x, y) \, dy \, dx = \int_c^d \int_a^b f(x, y) \, dx \, dy$$

- If $f(x, y) = g(x)h(y)$, then:

$$\iint_R f(x, y) \, dA = \left(\int_a^b g(x) \, dx \right) \times \left(\int_c^d h(y) \, dy \right)$$

General Regions

- **Type I Region** (bounded by functions of x):

$$D = \{(x, y) \mid a \leq x \leq b, g_1(x) \leq y \leq g_2(x)\}$$

$$\iint_D f(x, y) \, dA = \int_a^b \int_{g_1(x)}^{g_2(x)} f(x, y) \, dy \, dx$$

- **Type II Region** (bounded by functions of y):

$$D = \{(x, y) \mid c \leq y \leq d, h_1(y) \leq x \leq h_2(y)\}$$

$$\iint_D f(x, y) \, dA = \int_c^d \int_{h_1(y)}^{h_2(y)} f(x, y) \, dx \, dy$$

- **Area of D :** $A(D) = \iint_D 1 \, dA$.

Double Integrals in Polar Coordinates

- **Transformations:** $x = r \cos \theta, y = r \sin \theta, x^2 + y^2 = r^2$.
- **Area Element:** $dA = r \, dr \, d\theta$ (Jacobian is r).
- **Integral formula** over polar rectangle $R = [a, b] \times [\alpha, \beta]$:

$$\iint_R f(x, y) \, dA = \int_{\alpha}^{\beta} \int_a^b f(r \cos \theta, r \sin \theta) r \, dr \, d\theta$$

- **General Polar Region:** $D = \{(r, \theta) \mid \alpha \leq \theta \leq \beta, h_1(\theta) \leq r \leq h_2(\theta)\}$:

$$\iint_D f(x, y) \, dA = \int_{\alpha}^{\beta} \int_{h_1(\theta)}^{h_2(\theta)} f(r \cos \theta, r \sin \theta) r \, dr \, d\theta$$

8. Triple Integrals

Rectangular Box & Fubini's

- For $B = [a, b] \times [c, d] \times [r, s]$:

$$\iiint_B f(x, y, z) \, dV = \int_r^s \int_c^d \int_a^b f(x, y, z) \, dx \, dy \, dz$$

General Regions

- **Type 1:** $E = \{(x, y, z) \mid (x, y) \in D, u_1(x, y) \leq z \leq u_2(x, y)\}$ where D is the projection of E onto the xy -plane:

$$\iiint_E f(x, y, z) \, dV = \iint_D \left(\int_{u_1(x, y)}^{u_2(x, y)} f(x, y, z) \, dz \right) dA$$

- **Type 2:** $E = \{(x, y, z) \mid (y, z) \in D, u_1(y, z) \leq x \leq u_2(y, z)\}$ (D in yz -plane).
- **Type 3:** $E = \{(x, y, z) \mid (x, z) \in D, u_1(x, z) \leq y \leq u_2(x, z)\}$ (D in xz -plane).
- **Volume of E :** $V(E) = \iiint_E 1 \, dV$.

Cylindrical Coordinates

- **Transformations:** $x = r \cos \theta, y = r \sin \theta, z = z$.
- **Volume Element:** $dV = r \, dz \, dr \, d\theta$.
- **Integration Formula** over Type 1 Region E :

$$\iiint_E f(x, y, z) \, dV = \int_{\alpha}^{\beta} \int_{h_1(\theta)}^{h_2(\theta)} \int_{u_1(r \cos \theta, r \sin \theta)}^{u_2(r \cos \theta, r \sin \theta)} f(r \cos \theta, r \sin \theta, z) r \, dz \, dr \, d\theta$$

Spherical Coordinates

- **Transformations:**

$$x = \rho \sin \phi \cos \theta, \quad y = \rho \sin \phi \sin \theta, \quad z = \rho \cos \phi$$

where $\rho \geq 0, 0 \leq \phi \leq \pi$, and $x^2 + y^2 + z^2 = \rho^2$.

- **Volume Element:** $dV = \rho^2 \sin \phi \, d\rho \, d\theta \, d\phi$.
- **Spherical Wedge** $E = \{(\rho, \theta, \phi) \mid a \leq \rho \leq b, \alpha \leq \theta \leq \beta, c \leq \phi \leq d\}$:

$$\iiint_E f(x, y, z) \, dV = \int_c^d \int_{\alpha}^{\beta} \int_a^b f(\rho \sin \phi \cos \theta, \rho \sin \phi \sin \theta, \rho \cos \phi) \rho^2 \sin \phi \, d\rho \, d\theta \, d\phi$$